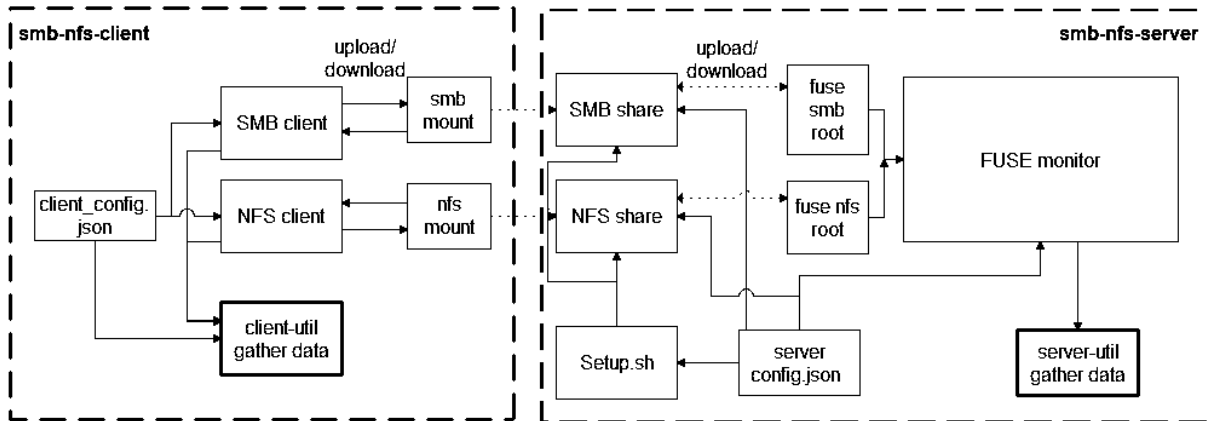




## 1. Config Server

Run **setup.sh**

```
ubuntu20@ubuntu: ~/Documents/smb-nfs-util
ubuntu20@ubuntu:~/Documents/smb-nfs-util$ sudo ./setup.sh
```

```
ubuntu20@ubuntu:~/Documents/smb-nfs-util$ sudo ./setup.sh
[sudo] password for ubuntu20:
Setting up the SMB/NFS server...
Get:1 file:/usr/lib/expressvpn/repo_mirror.list Mirrorlist [117 B]
Get:4 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:5 http://us.archive.ubuntu.com/ubuntu focal InRelease [265 kB]
Get:2 https://repo.expressvpn.com/public/deb/debian any-version InRelease [4,580 B]
Get:3 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Get:8 https://packages.microsoft.com/repos/code stable/main arm64 Packages [18.1 kB]
Get:9 https://packages.microsoft.com/repos/code stable/main amd64 Packages [17.9 kB]
Get:10 https://packages.microsoft.com/repos/code stable/main armhf Packages [18.1 kB]
Get:6 https://repo.expressvpn.com/public/deb/debian any-version/main i386 Packages [812 B]
Get:11 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,126 kB]
Get:7 https://repo.expressvpn.com/public/deb/debian any-version/main amd64 Packages [698 B]
Get:12 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:13 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:14 http://us.archive.ubuntu.com/ubuntu focal/main i386 Packages [718 kB]
Get:15 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [797 kB]
Get:16 http://us.archive.ubuntu.com/ubuntu focal/main amd64 Packages [970 kB]
Get:17 http://us.archive.ubuntu.com/ubuntu focal/main Translation-en [506 kB]
Get:18 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [464 kB]
Get:19 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [65.3 kB]
Get:20 http://security.ubuntu.com/ubuntu focal-security/main DEP-11 48x48 Icons [24.2 kB]
Get:21 http://security.ubuntu.com/ubuntu focal-security/main DEP-11 64x64 Icons [42.9 kB]
Get:22 http://security.ubuntu.com/ubuntu focal-security/main DEP-11 64x64@2 Icons [29 B]
```

```
Get:130 http://us.archive.ubuntu.com/ubuntu focal-backports/multiverse amd64 c-n-f Metadata [116 B]
Fetched 60.8 MB in 11s (5,730 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree
Reading state information... Done
fuse is already the newest version (2.9.9-3).
nfs-common is already the newest version (1:1.3.4-2.5ubuntu3.7).
nfs-kernel-server is already the newest version (1:1.3.4-2.5ubuntu3.7).
samba is already the newest version (2:4.15.13+dfsg-0ubuntu0.20.04.7).
smbclient is already the newest version (2:4.15.13+dfsg-0ubuntu0.20.04.7).
jq is already the newest version (1.6-1ubuntu0.20.04.1).
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
Reading package lists... Done
Building dependency tree
Reading state information... Done
python3 is already the newest version (3.8.2-0ubuntu2).
python3-pip is already the newest version (20.0.2-5ubuntu1.10).
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
Creating NFS export at /var/www/nfs_export
Creating SMB share at /smb_share
SMB share already exists in /etc/samba/smb.conf
NFS export already exists in /etc/exports
Setting RPCNFSDCOUNT in /etc/default/nfs-kernel-server
Setting NFSD_OPTS in /etc/default/nfs-kernel-server
```

```
python3-pip is already the newest version (20.0.2-5ubuntu1.10).
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
Creating NFS export at /var/www/nfs_export
Creating SMB share at /smb_share
SMB share already exists in /etc/samba/smb.conf
NFS export already exists in /etc/exports
Setting RPCNFSDCOUNT in /etc/default/nfs-kernel-server
Setting NFSD_OPTS in /etc/default/nfs-kernel-server
chmod: changing permissions of '/fuse_nfs_root': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/touchtest': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/test': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/testdfsdf': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/testfile.bin': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/testfile_1.bin': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/testd': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/ttestfiles': Operation not permitted
chmod: changing permissions of '/fuse_nfs_root/testdfsdfsdf': Operation not permitted
Mounting SMB share using FUSE
mount error(16): Device or resource busy
Refer to the mount.cifs(8) manual page (e.g. man mount.cifs) and kernel log messages (dmesg)
SMB share already exists in /etc/fstab
Mounting NFS export using FUSE
NFS export already exists in /etc/fstab
Requirement already satisfied: tqdm==4.66.4 in /usr/local/lib/python3.8/dist-packages (from -r requirements.txt (line 1)) (4.66.4)
Requirement already satisfied: pandas in /usr/local/lib/python3.8/dist-packages (from -r requirements.txt (line 2)) (2.0.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.8/dist-packages (from -r requirements.txt (line 3)) (1.3.2)
Requirement already satisfied: python-dateutil<=2.8.2 in /usr/local/lib/python3.8/dist-packages (from pandas->-r requirements.txt (line 2)) (2.9.0.post0)
Requirement already satisfied: numpy>=1.20.3; python_version < "3.10" in /usr/local/lib/python3.8/dist-packages (from pandas->-r requirements.txt (line 2)) (1.24.4)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.8/dist-packages (from pandas->-r requirements.txt (line 2)) (2024.1)
Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.8/dist-packages (from pandas->-r requirements.txt (line 2)) (2024.1)
Requirement already satisfied: joblib>=1.1.1 in /usr/local/lib/python3.8/dist-packages (from scikit-learn->-r requirements.txt (line 3)) (1.4.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in /usr/local/lib/python3.8/dist-packages (from scikit-learn->-r requirements.txt (line 3)) (3.4.0)
Requirement already satisfied: scipy>=1.5.0 in /usr/local/lib/python3.8/dist-packages (from scikit-learn->-r requirements.txt (line 3)) (1.10.1)
Requirement already satisfied: stx>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2->pandas->-r requirements.txt (line 2)) (1.14.0)
ubuntu20@ubuntu:~/Documents/smb-nfs-util$
```

The `setup.sh` script is a Bash script designed to set up an SMB/NFS server environment on a Linux system. Here's a breakdown of what the script does:

### 1. System Setup:

- The script begins by updating the package list and installing several system dependencies, including tools and services necessary for SMB (Samba) and NFS (Network File System) sharing, as well as Python3 and related tools.

### 2. SMB and NFS Configuration:

- The script reads configuration details from a `server_config.json` file, which contains settings for both NFS and SMB shares.
- It creates directories for both NFS exports and SMB shares, with full read/write permissions (`chmod -R 0777`).

### 3. SMB Share Setup:

- The script checks if the SMB share is already configured in the `/etc/samba/smb.conf` file. If not, it appends the necessary configuration for the SMB share to this file.

### 4. NFS Server Setup:

- The script starts the NFS kernel server and ensures the NFS export path is listed in the `/etc/exports` file. If not, it adds the export configuration.
- It also modifies `/etc/default/nfs-kernel-server` to set NFS-specific options like `RPCNFSDCOUNT`, `lease_time`, and `grace_time` using values from the `server_config.json`.

### 5. FUSE Setup for SMB and NFS:

- The script enables the `user_allow_other` option in `/etc/fuse.conf`, which allows non-root users to mount filesystems.
- It then creates directories for FUSE mounts for both SMB and NFS.
- The SMB and NFS shares are mounted using FUSE, and the script ensures these mounts are persistent across reboots by adding the appropriate entries to `/etc/fstab`.

### 6. Python Environment Setup:

- The script navigates to the directory where it's located and installs Python dependencies listed in a `requirements.txt` file.
- It concludes by running a Python script (`main.py`) located in the `src` directory.

Overall, this script automates the process of setting up a server environment that supports both SMB and NFS file sharing, along with some additional setup for a Python-based application. The configuration is mostly driven by the `server_config.json` file, which holds various paths and settings needed during the setup.

**Server\_Config.json**

```

server_config.json > {}
1  {
2      "smb": {
3          "server": "localhost",
4          "share_name": "/smb_share",
5          "username": "root",
6          "password": "root",
7          "client_machine_name": "",
8          "server_settings": {
9              "socket options": "TCP_NODELAY SO_RCVBUF=8192 SO_SNDBUF=8192",
10             "read raw": "yes",
11             "write raw": "yes",
12             "getwd cache": "yes",
13             "max xmit": 65535,
14             "dead time": 15,
15             "use sendfile": "yes",
16             "aio read size": 16384,
17             "aio write size": 16384,
18             "aio write behind": true
19         }
20     },
21     "nfs": {
22         "server": "localhost",
23         "export_path": "/var/www/nfs_export",
24         "server_settings": {
25             "rpcnfsdcount": 16,
26             "lease_time": 90,
27             "grace_time": 90
28         }
29     },
30     "fuse" : {
31         "fuse_smb_mount_point": "/fuse_smb",
32         "fuse_smb_root": "/fuse_smb_root",
33         "fuse_nfs_mount_point": "/fuse_nfs",
34         "fuse_nfs_root": "/fuse_nfs_root"
35     }
36 }
37

```

The `server\_config.json` file contains configuration settings for both the SMB and NFS server setups. Here's an explanation of the key components:

### ### SMB Configuration:

- server: The address of the SMB server, set to `localhost` in this case.
- share\_name: The path to the SMB share, which is `/smb\_share`.
- username & password: Credentials (`root`/`root`) used for accessing the SMB share.
- client\_machine\_name: This is an optional field, left empty in this configuration.
- server\_settings: A set of parameters that tune the SMB server's performance and behavior:
  - socket options: Network options like `TCP\_NODELAY` and buffer sizes (`SO\_RCVBUF`, `SO\_SNDBUF`).
  - read raw & write raw: Whether to use raw reads/writes for SMB.
  - getwd cache: Caching for directory lookups.
  - max xmit: Maximum transmission size set to `65535` bytes.
  - dead time: The number of minutes of inactivity before a connection is considered dead, set to `15`.

- use sendfile: Optimization for sending files.
- aio read/write size: Size for asynchronous I/O operations.
- aio write behind: Indicates whether write-behind is enabled for asynchronous I/O.

#### ### NFS Configuration:

- server: The address of the NFS server, also set to `"localhost"`.
- export\_path: The path where the NFS export will be located, set to `/var/www/nfs_export`.
- server\_settings: Additional settings for the NFS server:
  - rpcnfsdcount: Number of NFS daemons, set to `16`.
  - lease\_time & grace\_time: Lease and grace times for NFS, both set to `90` seconds.

#### ### FUSE Configuration:

- fuse\_smb\_mount\_point: The mount point for the SMB share when using FUSE, set to `/fuse_smb`.
- fuse\_smb\_root: The root directory for the SMB FUSE mount, set to `/fuse_smb_root`.
- fuse\_nfs\_mount\_point: The mount point for the NFS export when using FUSE, set to `/fuse_nfs`.
- fuse\_nfs\_root: The root directory for the NFS FUSE mount, set to `/fuse_nfs_root`.

#### ### Integration with `setup.sh`:

This JSON configuration file is referenced in the `setup.sh` script you provided. The script reads these settings to create and configure the necessary directories, set up the SMB and NFS shares, and mount them using FUSE. The values in this file, such as paths and server settings, dictate how the system is configured and how it will operate once the setup is complete.

Check NFS, SMB is mounted correctly

```

Requirement already satisfied: stx>=1.3 in /usr/lib/python3/dist-packages (from python-dateutil>=2.6.2)
ubuntu20@ubuntu:~/Documents/smb-nfs-util$ df -h
Filesystem                Size      Used Avail Use% Mounted on
udev                     1.9G         0   1.9G   0% /dev
tmpfs                    388M       3.5M   385M   1% /run
/dev/sda5                 39G       19G    19G  51% /
tmpfs                    1.9G         0   1.9G   0% /dev/shm
tmpfs                    5.0M       4.0K   5.0M   1% /run/lock
tmpfs                    1.9G         0   1.9G   0% /sys/fs/cgroup
/dev/loop0               128K      128K     0 100% /snap/bare/5
/dev/loop1               64M       64M     0 100% /snap/core20/2264
/dev/loop2               64M       64M     0 100% /snap/core20/2318
/dev/loop3               75M       75M     0 100% /snap/core22/1439
/dev/loop4              505M      505M     0 100% /snap/gnome-42-2204/172
/dev/loop6              347M      347M     0 100% /snap/gnome-3-38-2004/119
/dev/loop5               75M       75M     0 100% /snap/core22/1380
/dev/loop9               92M       92M     0 100% /snap/gtk-common-themes/1535
/dev/loop8              506M      506M     0 100% /snap/gnome-42-2204/176
/dev/loop11              13M       13M     0 100% /snap/snap-store/959
/dev/loop10              13M       13M     0 100% /snap/snap-store/1113
/dev/loop7              350M      350M     0 100% /snap/gnome-3-38-2004/143
/dev/loop12              39M       39M     0 100% /snap/snapd/21465
/dev/loop13              39M       39M     0 100% /snap/snapd/21759
/dev/sda1                511M       4.0K   511M   1% /boot/efi
localhost:/var/www/nfs_export 39G       19G    19G  51% /fuse_nfs_root
tmpfs                    388M       48K   388M   1% /run/user/1000
//localhost/smb_share     39G       21G    19G  53% /fuse_smb_root
ubuntu20@ubuntu:~/Documents/smb-nfs-util$

```

## Run Fuse Monitor

```
>> python3 server_fuse_monitor.py
```

```

ubuntu20@ubuntu:~/Documents/smb-nfs-util$ sudo python3 server_fuse_monitor.py
Mounting FUSE filesystem at /fuse_smb, root: /fuse_smb_root
Mounting FUSE filesystem at /fuse_nfs, root: /fuse_nfs_root

```

The `server\_fuse\_monitor.py` script is a Python program designed to monitor file I/O operations on SMB and NFS shares mounted using FUSE (Filesystem in Userspace). Here's a breakdown of what the script does:

### ### Key Components:

#### 1. Imports and Logging Setup:

- The script starts by importing necessary libraries, including `os`, `json`, `sys`, `errno`, `logging`, `pyinotify`, and `fuse`.
- Logging is configured to record messages in `logs/fuse\_utils.log`.
- A CSV file (`collected\_data/IO\_events.csv`) is set up to store details of file operations such as file path, size, event type (e.g., READ, WRITE), and timestamp.

#### 2. CSV Writing Function:

- The `write\_to\_csv` function logs the details of each file operation to the CSV file, including the file path, size, event type, and timestamp.

#### 3. FuseMonitor Class:

- This class extends `Operations` from the `fuse` module and overrides several methods to handle file operations:

- `getattr`: Retrieves file attributes (similar to `stat` in UNIX).
- `readdir`: Reads the contents of a directory.
- `read`: Handles file read operations, logging the event and writing details to the CSV.
- `write`: Handles file write operations, logging the event and writing details to the CSV.
- `truncate`: Truncates a file to a specified length.
- `open`, `release`, `create`, `unlink`: Manage opening, closing, creating, and deleting files, respectively.
- Each method logs the operation and writes to the CSV file for monitoring purposes.

#### 4. Main Function for FUSE Mount:

- The `main` function mounts the FUSE filesystem at the specified mount point and root directory. It runs in the foreground with several options like `allow_other` to permit other users to access the filesystem.

#### 5. EventHandler Class:

- This class is used with `pyinotify` to watch for specific file events such as `IN_CLOSE_WRITE` (file closed after writing) and `IN_CLOSE_NOWRITE` (file closed without writing).
- When these events occur, the script logs the event and writes the details to the CSV file.

#### 6. Inotify Watch Function:

- The `start_inotify_watch` function sets up an inotify watch on specified paths to monitor file events. It uses `pyinotify` to handle these events and process them using the `EventHandler` class.

#### 7. Main Execution Block:

- The script reads configuration settings from `server_config.json` to determine the paths for SMB and NFS shares.
- It then starts three processes:
  - An inotify watch process to monitor file events on the specified paths.
  - A FUSE filesystem process for the SMB share.
  - A FUSE filesystem process for the NFS export.
- These processes run concurrently to ensure that file operations are monitored and logged in real-time.

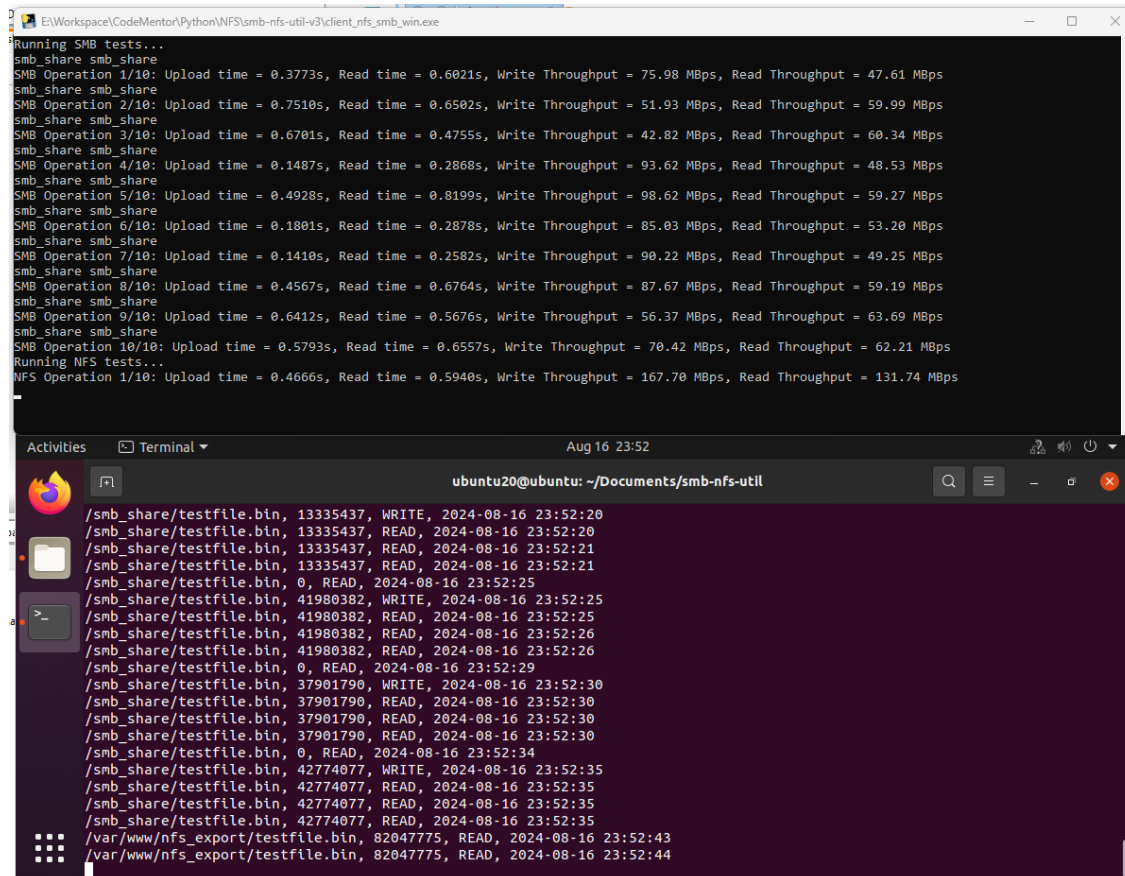
#### ### Purpose:

This script is designed to provide real-time monitoring and logging of file operations (reads, writes, truncations, etc.) on SMB and NFS shares. It creates a FUSE-based virtual filesystem that intercepts file operations and logs them to both a log file and a CSV file for further analysis. This setup is particularly useful for tracking file I/O activities in a networked environment where SMB and NFS shares are commonly used.

## Run Client

### 1. Windows

Run client\_nfs\_smb\_win.exe in windows

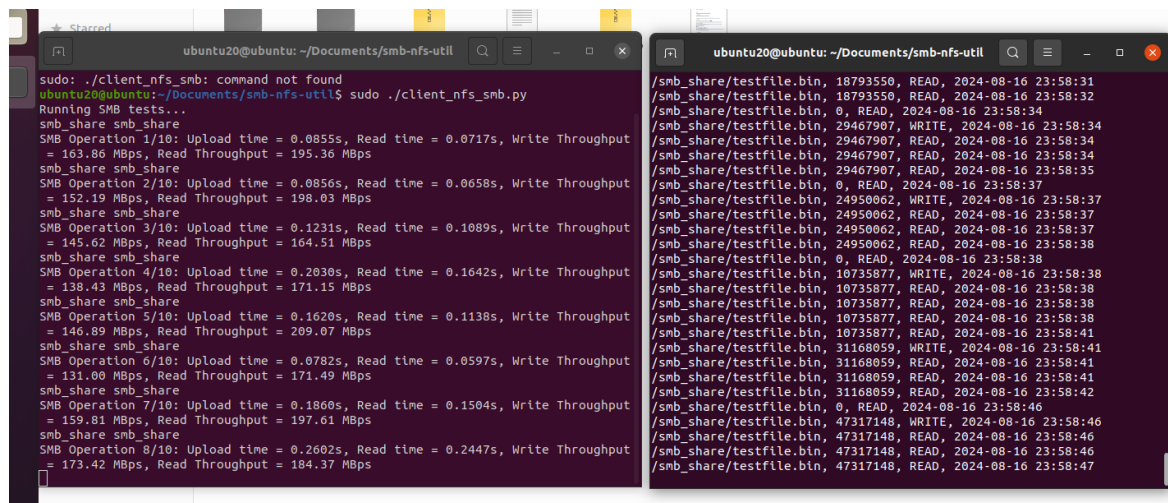


The image shows two terminal windows. The top window is a Windows command prompt running 'client\_nfs\_smb\_win.exe'. It displays SMB test results for 10 operations, showing upload and read times, write throughput, and read throughput. The bottom window is a Linux terminal running 'smb-nfs-util' and 'client\_nfs\_smb.py'. It displays SMB test results for 10 operations, showing upload and read times, write throughput, and read throughput. The Linux terminal also shows NFS test results for 10 operations, showing upload and read times, write throughput, and read throughput.

```
Running SMB tests...
smb_share smb_share
SMB Operation 1/10: Upload time = 0.3773s, Read time = 0.6021s, Write Throughput = 75.98 MBps, Read Throughput = 47.61 MBps
smb_share smb_share
SMB Operation 2/10: Upload time = 0.7510s, Read time = 0.6502s, Write Throughput = 51.93 MBps, Read Throughput = 59.99 MBps
smb_share smb_share
SMB Operation 3/10: Upload time = 0.6701s, Read time = 0.4755s, Write Throughput = 42.82 MBps, Read Throughput = 60.34 MBps
smb_share smb_share
SMB Operation 4/10: Upload time = 0.1487s, Read time = 0.2868s, Write Throughput = 93.62 MBps, Read Throughput = 48.53 MBps
smb_share smb_share
SMB Operation 5/10: Upload time = 0.4928s, Read time = 0.8199s, Write Throughput = 98.62 MBps, Read Throughput = 59.27 MBps
smb_share smb_share
SMB Operation 6/10: Upload time = 0.1801s, Read time = 0.2878s, Write Throughput = 85.03 MBps, Read Throughput = 53.20 MBps
smb_share smb_share
SMB Operation 7/10: Upload time = 0.1410s, Read time = 0.2582s, Write Throughput = 90.22 MBps, Read Throughput = 49.25 MBps
smb_share smb_share
SMB Operation 8/10: Upload time = 0.4567s, Read time = 0.6764s, Write Throughput = 87.67 MBps, Read Throughput = 59.19 MBps
smb_share smb_share
SMB Operation 9/10: Upload time = 0.6412s, Read time = 0.5676s, Write Throughput = 56.37 MBps, Read Throughput = 63.69 MBps
smb_share smb_share
SMB Operation 10/10: Upload time = 0.5793s, Read time = 0.6557s, Write Throughput = 70.42 MBps, Read Throughput = 62.21 MBps
Running NFS tests...
NFS Operation 1/10: Upload time = 0.4666s, Read time = 0.5940s, Write Throughput = 167.70 MBps, Read Throughput = 131.74 MBps
```

```
ubuntu20@ubuntu: ~/Documents/smb-nfs-util
/smb_share/testfile.bin, 13335437, WRITE, 2024-08-16 23:52:20
/smb_share/testfile.bin, 13335437, READ, 2024-08-16 23:52:20
/smb_share/testfile.bin, 13335437, READ, 2024-08-16 23:52:21
/smb_share/testfile.bin, 13335437, READ, 2024-08-16 23:52:21
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:52:25
/smb_share/testfile.bin, 41980382, WRITE, 2024-08-16 23:52:25
/smb_share/testfile.bin, 41980382, READ, 2024-08-16 23:52:25
/smb_share/testfile.bin, 41980382, READ, 2024-08-16 23:52:26
/smb_share/testfile.bin, 41980382, READ, 2024-08-16 23:52:26
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:52:29
/smb_share/testfile.bin, 37901790, WRITE, 2024-08-16 23:52:30
/smb_share/testfile.bin, 37901790, READ, 2024-08-16 23:52:30
/smb_share/testfile.bin, 37901790, READ, 2024-08-16 23:52:30
/smb_share/testfile.bin, 37901790, READ, 2024-08-16 23:52:30
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:52:34
/smb_share/testfile.bin, 42774077, WRITE, 2024-08-16 23:52:35
/smb_share/testfile.bin, 42774077, READ, 2024-08-16 23:52:35
/smb_share/testfile.bin, 42774077, READ, 2024-08-16 23:52:35
/smb_share/testfile.bin, 42774077, READ, 2024-08-16 23:52:35
/var/www/nfs_export/testfile.bin, 82047775, READ, 2024-08-16 23:52:43
/var/www/nfs_export/testfile.bin, 82047775, READ, 2024-08-16 23:52:44
```

### 2. Linux



The image shows a Linux terminal window running 'client\_nfs\_smb.py'. It displays SMB test results for 10 operations, showing upload and read times, write throughput, and read throughput. The terminal also shows NFS test results for 10 operations, showing upload and read times, write throughput, and read throughput.

```
sudo: ./client_nfs_smb: command not found
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ sudo ./client_nfs_smb.py
Running SMB tests...
smb_share smb_share
SMB Operation 1/10: Upload time = 0.0855s, Read time = 0.0717s, Write Throughput = 163.86 MBps, Read Throughput = 195.36 MBps
smb_share smb_share
SMB Operation 2/10: Upload time = 0.0856s, Read time = 0.0658s, Write Throughput = 152.10 MBps, Read Throughput = 198.03 MBps
smb_share smb_share
SMB Operation 3/10: Upload time = 0.1231s, Read time = 0.1089s, Write Throughput = 145.62 MBps, Read Throughput = 164.51 MBps
smb_share smb_share
SMB Operation 4/10: Upload time = 0.2030s, Read time = 0.1642s, Write Throughput = 138.43 MBps, Read Throughput = 171.15 MBps
smb_share smb_share
SMB Operation 5/10: Upload time = 0.1620s, Read time = 0.1138s, Write Throughput = 146.89 MBps, Read Throughput = 209.07 MBps
smb_share smb_share
SMB Operation 6/10: Upload time = 0.0782s, Read time = 0.0597s, Write Throughput = 131.00 MBps, Read Throughput = 171.49 MBps
smb_share smb_share
SMB Operation 7/10: Upload time = 0.1860s, Read time = 0.1504s, Write Throughput = 159.81 MBps, Read Throughput = 197.61 MBps
smb_share smb_share
SMB Operation 8/10: Upload time = 0.2602s, Read time = 0.2447s, Write Throughput = 173.42 MBps, Read Throughput = 184.37 MBps
```

```
/smb_share/testfile.bin, 18793550, READ, 2024-08-16 23:58:31
/smb_share/testfile.bin, 18793550, READ, 2024-08-16 23:58:32
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:58:34
/smb_share/testfile.bin, 29467907, WRITE, 2024-08-16 23:58:34
/smb_share/testfile.bin, 29467907, READ, 2024-08-16 23:58:34
/smb_share/testfile.bin, 29467907, READ, 2024-08-16 23:58:34
/smb_share/testfile.bin, 29467907, READ, 2024-08-16 23:58:35
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:58:37
/smb_share/testfile.bin, 24950062, WRITE, 2024-08-16 23:58:37
/smb_share/testfile.bin, 24950062, READ, 2024-08-16 23:58:37
/smb_share/testfile.bin, 24950062, READ, 2024-08-16 23:58:37
/smb_share/testfile.bin, 24950062, READ, 2024-08-16 23:58:38
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:58:38
/smb_share/testfile.bin, 10735877, WRITE, 2024-08-16 23:58:38
/smb_share/testfile.bin, 10735877, READ, 2024-08-16 23:58:38
/smb_share/testfile.bin, 10735877, READ, 2024-08-16 23:58:38
/smb_share/testfile.bin, 10735877, READ, 2024-08-16 23:58:38
/smb_share/testfile.bin, 10735877, READ, 2024-08-16 23:58:41
/smb_share/testfile.bin, 31168059, WRITE, 2024-08-16 23:58:41
/smb_share/testfile.bin, 31168059, READ, 2024-08-16 23:58:41
/smb_share/testfile.bin, 31168059, READ, 2024-08-16 23:58:41
/smb_share/testfile.bin, 31168059, READ, 2024-08-16 23:58:42
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:58:46
/smb_share/testfile.bin, 47317148, WRITE, 2024-08-16 23:58:46
/smb_share/testfile.bin, 47317148, READ, 2024-08-16 23:58:46
/smb_share/testfile.bin, 47317148, READ, 2024-08-16 23:58:46
```

The `client_nfs_smb.py` script is a Python program designed to test and measure the performance of file operations (upload and download) on SMB (Server Message Block) and NFS (Network File System) shares. It performs these operations by mounting the NFS share, uploading and downloading files, and then recording the throughput and latency in a CSV file.

### Key Components:

#### 1. Imports and Setup:

- The script imports several modules such as `os`, `platform`, `time`, `datetime`, `json`, `random`, `string`, `subprocess`, `csv`, and `logging`.
- It sets up logging to record messages in `logs/client_nfs_smb.log`.
- It also ensures that necessary directories (`logs` and `collected_data`) exist, creating them if they do not.

#### 2. CSV File Setup:

- The script prepares a CSV file (`collected_data/client_throughput.csv`) to store the results of the file operations, including the mode (SMB or NFS), file path, event type (upload or read), timestamp, read/write sizes, NFS version, TCP usage, file size in MB, throughput in Mbps, and latency in milliseconds.

#### 3. Helper Functions:

- **write\_to\_csv:** Writes the results of file operations to the CSV file.
- **generate\_random\_data:** Generates random data of a specified size for testing file uploads.
- **create\_test\_file:** Creates a temporary test file with random data.
- **smb\_upload\_read\_test:** Handles the upload and download of files to and from an SMB share, calculating the throughput and latency for each operation.
- **nfs\_upload\_read\_test:** Similar to `smb_upload_read_test`, but for NFS shares.
- **mount\_nfs:** Mounts the NFS share on the local system using the specified configuration options (e.g., NFS version, TCP/UDP, read/write sizes).
- **unmount\_nfs:** Unmounts the NFS share after the tests are complete.

#### 4. Main Execution Flow:

- The script reads configuration settings from a `client_config.json` file, which specifies the SMB and NFS configurations, as well as the number of operations to perform and the file path for the test file.
- **SMB Testing:**
  - If SMB is enabled in the configuration, the script creates an SMB client and runs a series of upload and download tests.
  - The results of each test, including throughput and latency, are logged and recorded in the CSV file.
- **NFS Testing:**
  - If NFS is enabled, the script mounts the NFS share and runs similar upload and download tests.
  - It also logs and records the results in the CSV file.

- After completing the tests, the script unmounts the NFS share.

## 5. Execution:

- The main() function coordinates the overall process, starting the SMB and NFS tests as configured.
- If both SMB and NFS are enabled, the script will perform tests on both systems and compare their performance.

### Purpose:

The script is designed to automate the testing of file upload and download operations on SMB and NFS shares. By measuring the throughput and latency for these operations, it helps to evaluate the performance of these network file systems under various conditions. The results are stored in a CSV file for analysis, making it easier to compare the performance of SMB and NFS and identify any potential bottlenecks or issues.

### Client\_config.json

```
{ } client_config.json > ...
1  {
2      "smb": {
3          "enabled": true,
4          "server": "192.168.220.129",
5          "share_name": "smb_share",
6          "username": "",
7          "password": "",
8          "domain_name": "",
9          "client_machine_name": "ClientMachine"
10     },
11     "nfs": {
12         "enabled": true,
13         "server": "192.168.220.129",
14         "remote_path": "/var/www/nfs_export",
15         "windows_local_mount_point": "Z:",
16         "linux_local_mount_point": "/nfs_export",
17         "client_settings": {
18             "nfsvers": "4",
19             "tcp": true,
20             "rsize": 65536,
21             "wsize": 65536
22         }
23     },
24     "operations": {
25         "num_operations": 10,
26         "file_path": "testfile.bin"
27     }
28 } ✨
29
```

The client\_config.json file provides configuration details for the client\_nfs\_smb.py script to perform tests on both SMB and NFS shares. Here's a breakdown of the key elements in this configuration file:

### Configuration Sections:

#### 1. SMB Configuration (smb):

- **enabled:** Set to true, indicating that SMB testing is enabled.
- **server:** The IP address of the SMB server, in this case, 192.168.220.129.
- **share\_name:** The name of the SMB share, which is smb\_share.
- **username & password:** These fields are currently empty, meaning no specific credentials are provided.
- **domain\_name:** Also left empty; this field would typically be used if the SMB server is part of a domain.
- **client\_machine\_name:** The name of the client machine accessing the SMB share, set to "ClientMachine".

#### 2. NFS Configuration (nfs):

- **enabled:** Set to true, indicating that NFS testing is enabled.
- **server:** The IP address of the NFS server, which is also 192.168.220.129.
- **remote\_path:** The remote path on the NFS server, /var/www/nfs\_export, which is the directory to be accessed.
- **windows\_local\_mount\_point:** The local mount point for the NFS share on a Windows system, set to "Z:".
- **linux\_local\_mount\_point:** The local mount point for the NFS share on a Linux system, set to "/nfs\_export".
- **client\_settings:** Additional settings for the NFS client:
  - **nfsvers:** Specifies the NFS version, set to "4".
  - **tcp:** Set to true, meaning the NFS communication will use TCP instead of UDP.
  - **rsize:** The read buffer size, set to 65536 bytes (64 KB).
  - **wsiz:** The write buffer size, also set to 65536 bytes (64 KB).

#### 3. Operations Configuration (operations):

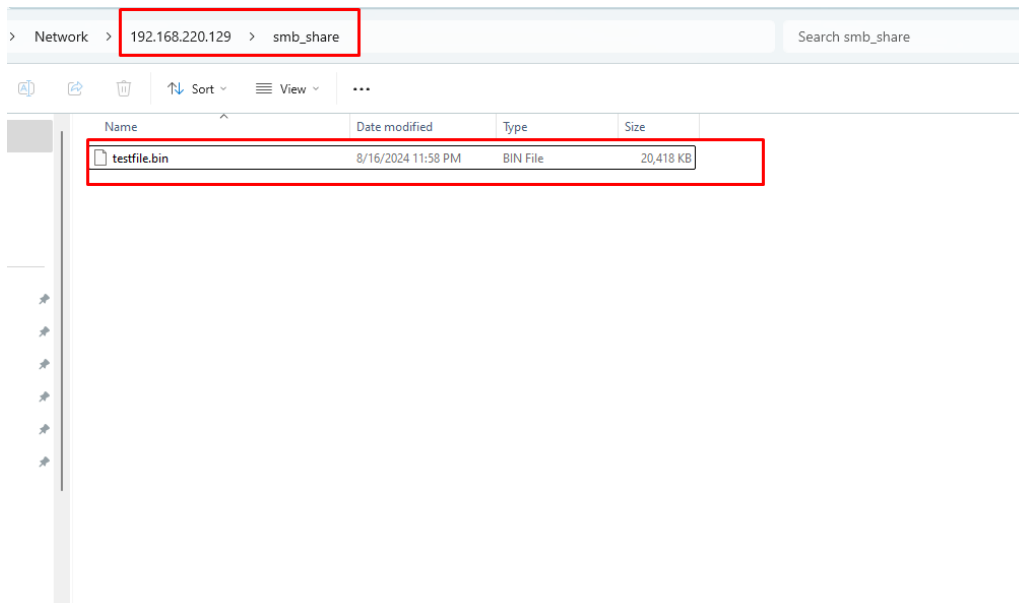
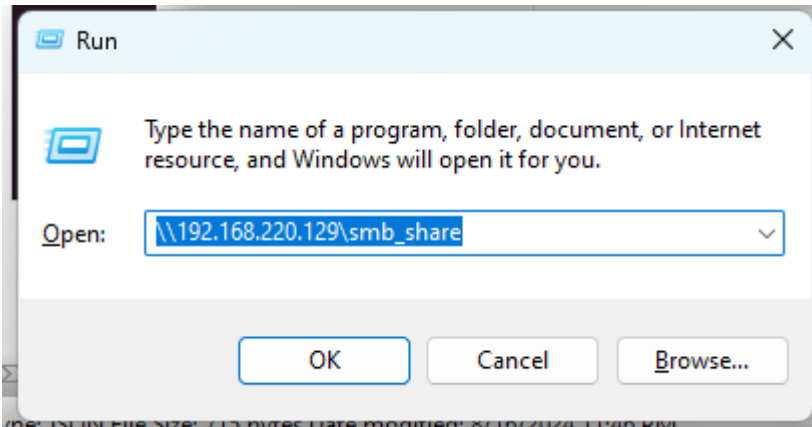
- **num\_operations:** Specifies the number of file operations (uploads/downloads) to perform, set to 10.
- **file\_path:** The file path or name for the test file, set to "testfile.bin".

### Purpose:

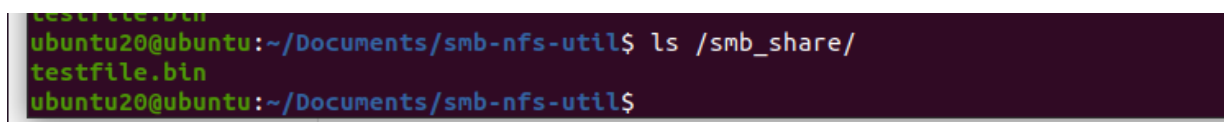
This configuration file directs the client\_nfs\_smb.py script to perform 10 file operations (uploads and downloads) on both the SMB and NFS shares hosted on the server at 192.168.220.129. It provides the necessary details for the script to connect to the shares, including server addresses, share names, and client-specific settings like buffer sizes and NFS version. The results of these operations will be used to assess the performance of both SMB and NFS under the specified conditions.

## How to check smb\_share connection in Windows

Open Run



## How to check smb\_share connection in Linux



## How to check server is monitoring SMB and NFS share

```
ubuntu20@ubuntu: ~/Documents/smb-nfs-util
/smb_share/testfile.bin, 16820754, READ, 2024-08-16 23:58:49
/smb_share/testfile.bin, 0, READ, 2024-08-16 23:58:50
/smb_share/testfile.bin, 20907484, WRITE, 2024-08-16 23:58:50
/smb_share/testfile.bin, 20907484, READ, 2024-08-16 23:58:50
/smb_share/testfile.bin, 20907484, READ, 2024-08-16 23:58:50
/smb_share/testfile.bin, 20907484, READ, 2024-08-16 23:58:51
/var/www/nfs_export/testfile.bin, 64076540, WRITE, 2024-08-16 23:58:56
/var/www/nfs_export/testfile.bin, 64076540, READ, 2024-08-16 23:58:56
/var/www/nfs_export/testfile.bin, 74363347, WRITE, 2024-08-16 23:59:04
/var/www/nfs_export/testfile.bin, 74363347, READ, 2024-08-16 23:59:04
/var/www/nfs_export/testfile.bin, 87848585, WRITE, 2024-08-16 23:59:12
/var/www/nfs_export/testfile.bin, 87848585, READ, 2024-08-16 23:59:12
/var/www/nfs_export/testfile.bin, 84061006, WRITE, 2024-08-16 23:59:20
/var/www/nfs_export/testfile.bin, 84061006, READ, 2024-08-16 23:59:20
/var/www/nfs_export/testfile.bin, 61891425, WRITE, 2024-08-16 23:59:26
/var/www/nfs_export/testfile.bin, 61891425, READ, 2024-08-16 23:59:26
/var/www/nfs_export/testfile.bin, 55263744, WRITE, 2024-08-16 23:59:31
/var/www/nfs_export/testfile.bin, 55263744, READ, 2024-08-16 23:59:31
/var/www/nfs_export/testfile.bin, 15248216, READ, 2024-08-16 23:59:33
/var/www/nfs_export/testfile.bin, 15248216, WRITE, 2024-08-16 23:59:33
/var/www/nfs_export/testfile.bin, 70124563, WRITE, 2024-08-16 23:59:39
/var/www/nfs_export/testfile.bin, 70124563, READ, 2024-08-16 23:59:39
/var/www/nfs_export/testfile.bin, 54504881, WRITE, 2024-08-16 23:59:44
/var/www/nfs_export/testfile.bin, 54504881, READ, 2024-08-16 23:59:44
/var/www/nfs_export/testfile.bin, 44505819, WRITE, 2024-08-16 23:59:48
/var/www/nfs_export/testfile.bin, 44505819, READ, 2024-08-16 23:59:48
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:03:07
```

In server.logs has each file I/O in SMB share and NFS export

So server monitor every file operations in SMB and NFS share and FUST root.

```
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /smb_share/
testfile.bin
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls
client_config.json  logs  __pycache__  server_config.json
collected_data     nmb   pyinotify.py  server_fuse_monitor.py
fuse.py             performance_analyzer.py  README.MD  setup.sh
inotify             pyasn1  recommendations.txt  smb
requirements.txt    smb_client.py
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ cp README.MD /fuse
fuse_mount/ fuse_nfs/ fuse_nfs_root/ fuse_smb/ fuse_smb_root/
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ cp README.MD /fuse
fuse_mount/ fuse_nfs/ fuse_nfs_root/ fuse_smb/ fuse_smb_root/
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ cp README.MD /fuse_smb
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /fuse_smb
README.MD testfile.bin
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$
```

```
/var/www/nfs_export/testfile.bin, 44505819, READ, 2024-08-16 23:59:48
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:03:07
/smb_share/README.MD, 0, READ, 2024-08-17 00:11:53
Write: /README.MD (size: 180, offset: 0)
/README.MD, 180, WRITE, 2024-08-17 00:11:53
/smb_share/README.MD, 180, READ, 2024-08-17 00:11:54
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:11:54
/smb_share/README.MD, 180, WRITE, 2024-08-17 00:11:54
/smb_share/README.MD, 180, READ, 2024-08-17 00:11:54
/smb_share/README.MD, 180, WRITE, 2024-08-17 00:11:54
/smb_share/README.MD, 180, READ, 2024-08-17 00:11:54
/smb_share/README.MD, 180, READ, 2024-08-17 00:11:54
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:11:54
/smb_share/README.MD, 180, READ, 2024-08-17 00:12:07
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:12:07
```

```
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /fuse_smb
README.MD testfile.bin
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ cp README.MD /fuse_nfs
README.MD testfile.bin
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /fuse_nfs
README.MD testd testdfsd testdfsd1 testfile.i.bin touchtest
test testdfsd testfile.i0.bin testfile.bin ttestfiles
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$
```

```
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:11:54
/smb_share/README.MD, 180, READ, 2024-08-17 00:12:07
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:12:07
[[Write: /README.MD (size: 180, offset: 0)
/README.MD, 180, WRITE, 2024-08-17 00:12:50
/var/www/nfs_export/README.MD, 180, WRITE, 2024-08-17 00:12:50
/var/www/nfs_export/README.MD, 180, WRITE, 2024-08-17 00:14:47
/smb_share/README.MD, 180, READ, 2024-08-17 00:17:59
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:17:59
```

Fuse\_smb

```
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ cp README.MD /fuse_smb
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /fuse_smb
README.MD testfile.bin
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ cp README.MD /fuse_nfs
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /fuse_nfs
README.MD testd testdfsd testdfsd1 testfile.i.bin touchtest
test testdfsd testfile.i0.bin testfile.bin ttestfiles
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$ ls /fuse_smb_root/
README.MD testfile.bin
ubuntu20@ubuntu: ~/Documents/smb-nfs-util$
```

```
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:11:54
/smb_share/README.MD, 180, READ, 2024-08-17 00:12:07
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:12:07
[[Write: /README.MD (size: 180, offset: 0)
/README.MD, 180, WRITE, 2024-08-17 00:12:50
/var/www/nfs_export/README.MD, 180, WRITE, 2024-08-17 00:12:50
/var/www/nfs_export/README.MD, 180, WRITE, 2024-08-17 00:14:47
/smb_share/README.MD, 180, READ, 2024-08-17 00:17:59
/smb_share/testfile.bin, 20907484, READ, 2024-08-17 00:17:59
```

## Collected Data

When client and server running. All I/O operations are logged in csv to be used for machine learning

### - Client Data

|    | A    | B            | C          | D               | E     | F     | G       | H    | I       | J               | K          |
|----|------|--------------|------------|-----------------|-------|-------|---------|------|---------|-----------------|------------|
| 1  | mode | filepath     | event_type | timestamp       | rsize | wsize | nfsvers | tcp  | size_MB | throughput_Mbps | latency_ms |
| 2  | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 29.97   | 83.4            | 359.3      |
| 3  | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 29.97   | 51.25           | 584.6      |
| 4  | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 41.27   | 61.63           | 669.6      |
| 5  | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 41.27   | 57.47           | 718        |
| 6  | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 32.42   | 45.57           | 711.3      |
| 7  | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 32.42   | 55.71           | 581.9      |
| 8  | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 34.24   | 52.26           | 655.1      |
| 9  | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 34.24   | 58.56           | 584.6      |
| 10 | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 32.57   | 72.3            | 450.5      |
| 11 | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 32.57   | 56.27           | 578.8      |
| 12 | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 14.07   | 74.49           | 188.9      |
| 13 | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 14.07   | 55.2            | 254.9      |
| 14 | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 30.24   | 67.67           | 446.8      |
| 15 | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 30.24   | 60.81           | 497.3      |
| 16 | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 15.43   | 79.67           | 193.6      |
| 17 | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 15.43   | 59.78           | 258        |
| 18 | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 16.72   | 41.39           | 404.1      |
| 19 | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 16.72   | 62.58           | 267.2      |
| 20 | smb  | testfile.bin | upload     | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 22.69   | 67.09           | 338.2      |
| 21 | smb  | testfile.bin | read       | 8/16/2024 23:48 | 0     | 0     | 0       | 0    | 22.69   | 55.62           | 408        |
| 22 | nfs  | testfile.bin | upload     | 8/16/2024 23:48 | 65536 | 65536 | 4       | TRUE | 52.16   | 190.38          | 274        |
| 23 | nfs  | testfile.bin | read       | 8/16/2024 23:48 | 65536 | 65536 | 4       | TRUE | 52.16   | 105.14          | 496.1      |
| 24 | nfs  | testfile.bin | upload     | 8/16/2024 23:48 | 65536 | 65536 | 4       | TRUE | 22.37   | 193.96          | 115.3      |
| 25 | nfs  | testfile.bin | read       | 8/16/2024 23:48 | 65536 | 65536 | 4       | TRUE | 22.37   | 135.56          | 165        |
| 26 | nfs  | testfile.bin | upload     | 8/16/2024 23:48 | 65536 | 65536 | 4       | TRUE | 11.26   | 195.98          | 57.5       |
| 27 | nfs  | testfile.bin | read       | 8/16/2024 23:48 | 65536 | 65536 | 4       | TRUE | 11.26   | 129.84          | 86.7       |

```
tnotlry pyasn1 requirements.txt smb_client.py
ubuntu20@ubuntu:~/Documents/smb-nfs-util$ cd collected_data/
ubuntu20@ubuntu:~/Documents/smb-nfs-util/collected_data$ ls
client_througput.csv IO_events.csv
ubuntu20@ubuntu:~/Documents/smb-nfs-util/collected_data$
```

```
client_througput.csv IO_events.csv
ubuntu20@ubuntu:~/Documents/smb-nfs-util/collected_data$ cat client_througput.csv
mode,filepath,event_type,timestamp,rsize,wsize,nfsvers,tcp,size_MB,throughput_Mbps,latency_ms
smb,testfile.bin,upload,2024-07-27 03:05:43,0,0,0,0,43.5,118.43,367.3
smb,testfile.bin,read,2024-07-27 03:05:43,0,0,0,0,43.5,189.94,229.0
smb,testfile.bin,upload,2024-07-27 03:05:44,0,0,0,0,10.88,123.5,88.1
smb,testfile.bin,read,2024-07-27 03:05:44,0,0,0,0,10.88,252.9,43.0
smb,testfile.bin,upload,2024-07-27 03:05:49,0,0,0,0,45.69,162.74,280.8
smb,testfile.bin,read,2024-07-27 03:05:49,0,0,0,0,45.69,280.72,162.8
smb,testfile.bin,upload,2024-07-27 03:05:51,0,0,0,0,19.25,152.34,126.4
smb,testfile.bin,read,2024-07-27 03:05:51,0,0,0,0,19.25,279.78,68.8
smb,testfile.bin,upload,2024-07-27 03:05:56,0,0,0,0,48.98,134.82,363.3
smb,testfile.bin,read,2024-07-27 03:05:56,0,0,0,0,48.98,176.72,277.1
smb,testfile.bin,upload,2024-07-27 03:05:57,0,0,0,0,10.79,119.88,90.0
smb,testfile.bin,read,2024-07-27 03:05:57,0,0,0,0,10.79,149.29,72.3
smb,testfile.bin,upload,2024-07-27 03:06:00,0,0,0,0,23.52,117.01,201.0
smb,testfile.bin,read,2024-07-27 03:06:00,0,0,0,0,23.52,154.44,152.29999999999998
smb,testfile.bin,upload,2024-07-27 03:06:03,0,0,0,0,28.38,122.22,232.2
smb,testfile.bin,read,2024-07-27 03:06:03,0,0,0,0,28.38,199.2,142.5
smb,testfile.bin,upload,2024-07-27 03:06:07,0,0,0,0,34.68,153.12,226.5
smb,testfile.bin,read,2024-07-27 03:06:07,0,0,0,0,34.68,249.62,138.9
smb,testfile.bin,upload,2024-07-27 03:06:12,0,0,0,0,43.81,144.98,302.20000000000005
smb,testfile.bin,read,2024-07-27 03:06:12,0,0,0,0,43.81,180.05,243.29999999999998
```

- Server data

```
ubuntu20@ubuntu:~/Documents/smb-nfs-util/collected_data$ cat IO_events.csv
filepath,size,event_type,timestamp
/smb_share/testfile.bin,0,READ,2024-07-29 21:08:58
/smb_share/testfile.bin,45482184,WRITE,2024-07-29 21:08:58
/smb_share/testfile.bin,45482184,READ,2024-07-29 21:08:58
/smb_share/testfile.bin,45482184,READ,2024-07-29 21:08:58
/smb_share/testfile.bin,0,READ,2024-07-29 21:09:02
/smb_share/testfile.bin,43014503,WRITE,2024-07-29 21:09:02
/smb_share/testfile.bin,43014503,READ,2024-07-29 21:09:02
/smb_share/testfile.bin,43014503,READ,2024-07-29 21:09:03
/smb_share/testfile.bin,0,READ,2024-07-29 21:09:06
/smb_share/testfile.bin,36608012,WRITE,2024-07-29 21:09:06
/smb_share/testfile.bin,36608012,READ,2024-07-29 21:09:06
/smb_share/testfile.bin,36608012,READ,2024-07-29 21:09:06
/smb_share/testfile.bin,0,READ,2024-07-29 21:09:10
/smb_share/testfile.bin,47829308,WRITE,2024-07-29 21:09:11
/smb_share/testfile.bin,47829308,READ,2024-07-29 21:09:11
/smb_share/testfile.bin,47829308,READ,2024-07-29 21:09:11
/smb_share/testfile.bin,0,READ,2024-07-29 21:09:13
/smb_share/testfile.bin,26990939,WRITE,2024-07-29 21:09:14
/smb_share/testfile.bin,26990939,READ,2024-07-29 21:09:14
/smb_share/testfile.bin,26990939,READ,2024-07-29 21:09:14
/smb_share/testfile.bin,0,READ,2024-07-29 21:09:15
```

Logs folder contains all logs